

베이지안 기계학습

7. Model selection

이경재

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Bayesian hypothesis test

Bayesian Hypothesis Test

- ▶ We want to test

$$H_0 : \theta \in \Theta_0 \quad \text{vs.} \quad H_1 : \theta \in \Theta_1, \quad \Theta = \Theta_0 \uplus \Theta_1$$

based on the observed data.

- ▶ H_0 : the null hypothesis
- ▶ H_1 : the alternative hypothesis
- ▶ We will impose priors

1. for each hypothesis: $\pi(H_0)$ and $\pi(H_1)$

2. for θ under each hypothesis: $\pi(\theta | H_0)$ and $\pi(\theta | H_1)$

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Priors: Step 1

$$\pi_0 = \pi(H_0) = \pi(\theta \in \Theta_0)$$

$$\pi_1 = \pi(H_1) = \pi(\theta \in \Theta_1)$$

- ▶ $\pi(H_0)$: the prior probability that H_0 is true
- ▶ $\pi(H_1)$: the prior probability that H_1 is true
- ▶ $\pi(H_0) + \pi(H_1) = 1$

Priors: Step 2

- Under each hypothesis, we use the following priors:

$$\theta \mid H_0 \sim g_0 \quad \text{such that } g_0(\Theta_0) = 1,$$

$$\theta \mid H_1 \sim g_1 \quad \text{such that } g_1(\Theta_1) = 1.$$

- By considering Step 1 and Step 2 in turn, it is equivalent to consider the following mixture prior:

$$\begin{aligned}\pi(\theta) &= \pi(H_0)\pi(\theta \mid H_0) + \pi(H_1)\pi(\theta \mid H_1) \\ &= \pi_0 g_0(\theta) + \pi_1 g_1(\theta).\end{aligned}$$

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Posterior

$$p_0 = \pi(H_0 \mid x) = \pi(\theta \in \Theta_0 \mid x)$$

$$p_1 = \pi(H_1 \mid x) = \pi(\theta \in \Theta_1 \mid x)$$

- ▶ $\pi(H_0 \mid x)$: the posterior probability that H_0 is true
- ▶ $\pi(H_1 \mid x)$: the posterior probability that H_1 is true

Bayesian Hypothesis Test

- ▶ For a Bayesian hypothesis test, we calculate the posterior probability for each model,

$$\pi(H_0 \mid x) \quad \text{and} \quad \pi(H_1 \mid x),$$

rather than the posterior for certain parameters.

- ▶ The interpretation is that a hypothesis with a higher posterior probability is considered more plausible.

Bayesian Hypothesis Test

- Note that

$$\begin{aligned}\pi(H_i | x) &\propto \pi(H_i)f(x | H_i) \\ &= \pi(H_i) \int_{\theta \in \Theta_i} f(x | \theta, H_i)\pi(\theta | H_i)d\theta, \quad i = 0, 1.\end{aligned}$$

- We focus on the cases where $f(x | \theta, H_i) = f(x | \theta)$.

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- We focus on the cases where $f(x | \theta, H_i) = f(x | \theta)$.

*Posterior

- How to calculate the posterior probabilities?

$$\begin{aligned}\pi(H_i | x) &= \frac{f(x, H_i)}{f(x)} \\ &= \frac{f(x, H_i)}{f(x, H_0) + f(x, H_1)}, \quad \text{for } i = 0, 1,\end{aligned}$$

where

$$\begin{aligned}f(x, H_i) &= \pi(H_i)f(x | H_i) \\ &= \pi(H_i) \int_{\Theta_i} f(x | \theta)\pi(\theta | H_i)d\theta.\end{aligned}$$

*Posterior Odds

- ▶ Hence the posterior odds is as follows:

$$\frac{\pi(H_0 | x)}{\pi(H_1 | x)} = \frac{\pi_0 \int f(x | \theta) g_0(\theta) d\theta}{\pi_1 \int f(x | \theta) g_1(\theta) d\theta}.$$

- ▶ Note that the above equation is equivalent to

$$\frac{\pi(H_0 | x)}{\pi(H_1 | x)} = \frac{\int_{\Theta_0} f(x | \theta) \pi(\theta) d\theta}{\int_{\Theta_1} f(x | \theta) \pi(\theta) d\theta}$$

because

$$g_i(\theta) = \frac{\pi(\theta)}{\pi_i} I(\theta \in \Theta_i) \quad \text{for } i = 0, 1.$$

Recall that $\pi(\theta) = \pi_0 g_0(\theta) + \pi_1 g_1(\theta)$.

Prior/Posterior Odds

- ▶ Prior odds can be understood as in favor of H_0 against H_1 :

$$\text{Prior odds} = \frac{\pi(H_0)}{\pi(H_1)}.$$

- ▶ Posterior odds can be understood as in favor of H_0 against H_1 :

$$\text{Posterior odds} = \frac{\pi(H_0 | x)}{\pi(H_1 | x)} = \frac{\pi(H_0) \int f(x | \theta) g_0(\theta) d\theta}{\pi(H_1) \int f(x | \theta) g_1(\theta) d\theta}.$$

Prior/Posterior Odds

- ▶ Therefore, the posterior odds depends on the prior odds.
- ▶ Since people may use different prior odds, it would be beneficial to somewhat mitigate the influence of prior odds in the posterior odds.

→ Bayes factor!

Prior/Posterior Odds

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- ▶ Since people may use different prior odds, it would be beneficial to somewhat mitigate the influence of prior odds in the posterior odds.
→ Bayes factor!

Bayes Factor

$$\begin{aligned} B_{01}(x) &= \frac{\pi(H_0 | x)}{\pi(H_1 | x)} \times \frac{\pi(H_1)}{\pi(H_0)} \\ &= \frac{\int f(x | \theta) g_0(\theta) d\theta}{\int f(x | \theta) g_1(\theta) d\theta} \\ &\equiv \frac{f(x | H_0)}{f(x | H_1)} \end{aligned}$$

- ▶ $f(x | H_i)$: the marginal density of x under H_i . (marginal likelihood)
- ▶ A Bayes factor is a ratio of marginal densities.
- ▶ If $\pi(H_0) = \pi(H_1)$, then Bayes Factor = posterior odds.

Bayes Factor

$$\begin{aligned} B_{01}(x) &= \frac{\pi(H_0 | x)}{\pi(H_1 | x)} \times \frac{\pi(H_1)}{\pi(H_0)} \\ \text{and } B_{10}(x) &= \frac{\pi(H_1 | x)}{\pi(H_0 | x)} \times \frac{\pi(H_0)}{\pi(H_1)} = \frac{1}{B_{01}(x)} \end{aligned}$$

► It is easy to see that

$$\pi(H_0 | x) = \frac{\pi(H_0)}{\pi(H_0) + \pi(H_1)B_{10}(x)}$$

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*Example: Simple vs. Simple

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1$$

► (Prior)

$$\begin{aligned} \pi_0 &= \pi(\theta = \theta_0) \quad \text{and} \quad \pi_1 = \pi(\theta = \theta_1) \\ \pi(\theta | H_0) &= \delta_{\theta_0}(\theta) \quad \text{and} \quad \pi(\theta | H_1) = \delta_{\theta_1}(\theta) \end{aligned}$$

► (Posterior)

$$\begin{aligned} \pi(\theta = \theta_0 | x) &= \frac{\pi_0 f(x | \theta_0)}{\pi_0 f(x | \theta_0) + \pi_1 f(x | \theta_1)} \\ \pi(\theta = \theta_1 | x) &= 1 - \pi(\theta = \theta_0 | x) \end{aligned}$$

► (Bayes factor)

$$B_{01}(x) = \frac{f(x | \theta_0)}{f(x | \theta_1)} \quad (\text{Likelihood ratio})$$

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Interpretation

- ▶ A Bayes Factor B_{10} greater than 1 supports H_1 over H_0 .
- ▶ Jeffreys provided the following interpretation:

$\log_{10} B_{10}(x)$	strength of the evidence for H_1
$0 \sim 0.5$	not worth than a bare mention
$0.5 \sim 1$	substantial
$1 \sim 2$	strong
> 2	decisive

Example

- ▶ Consider the Bernoulli model,

$$Y_i \mid \theta \stackrel{iid}{\sim} \text{Ber}(\theta), \quad i = 1, \dots, n,$$

and the hypothesis testing problem

$$H_0 : \theta = 0.5 \quad \text{vs} \quad H_1 : \theta \neq 0.5.$$

- ▶ Suppose we observe $x = \sum_{i=1}^n y_i = 115$, where $n = 200$.
- ▶ The likelihood is given by

$$f(x \mid \theta) = \binom{200}{115} \theta^{115} (1 - \theta)^{85}.$$

Example: Bayesian Hypothesis Test

- ▶ Note that $\pi(\theta | H_0) = \delta_{0.5}(\theta)$, so we have

$$\mathbb{P}(X = 115 | H_0) = f(x = 115 | \theta = 0.5) = \binom{200}{115} \left(\frac{1}{2}\right)^{200} = 0.005956.$$

- ▶ If we use $g_1(\theta) = \text{Unif}(\theta | 0, 1)$, then

$$\begin{aligned}\mathbb{P}(X = 115 | H_1) &= \int_0^1 f(x | \theta) g_1(\theta) d\theta \\ &= \int_0^1 \binom{200}{115} \theta^{115} (1 - \theta)^{85} d\theta = \frac{1}{201} = 0.004975.\end{aligned}$$

- ▶ The Bayes factor $B_{01}(x)$ is then 1.197, thus the evidence for H_0 is **barely worth mentioning** even though it points very slightly towards H_0 .

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*Example: Frequentist Hypothesis Test

- ▶ p-value:

$$Z \equiv \frac{X - 100}{\sqrt{50}} \sim N(0, 1)$$
$$\implies \mathbb{P}\left(|Z| \geq \frac{3}{\sqrt{2}}\right) \doteq 0.034$$

- ▶ A frequentist hypothesis test would reject H_0 at the 5% significance level.

Example: Remarks

- ▶ In this example, Bayesian hypothesis testing and frequentist hypothesis testing yield different results.
- ▶ Because these two hypothesis testing methods are based on different frameworks, they can provide different results even for the same data and model.
- ▶ A different choice of prior could result in a Bayes factor that is more in agreement with the frequentist hypothesis test.

*Example: Remarks

- ▶ A classical likelihood ratio test would have found the MLE for θ under H_1 , say $\hat{\theta}^{H_1} = \frac{115}{200} = 0.575$.
- ▶ That gives the likelihood ratio,

$$\frac{\mathbb{P}(X = 115 \mid \hat{\theta}^{H_0} = 0.5)}{\mathbb{P}(X = 115 \mid \hat{\theta}^{H_1} = 0.575)} = 0.1045,$$

and so pointing towards H_1 .

- ▶ H_1 is a more complex model than H_0 because it has a free parameter θ which allows it to model the data more closely.

*Example: Remarks

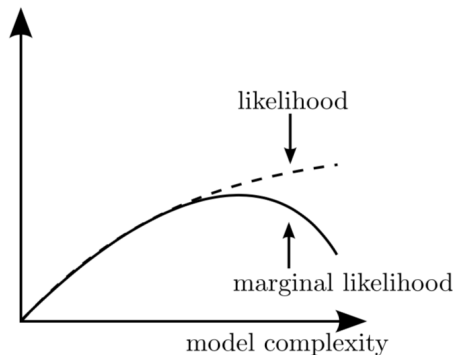


Figure: From Michael I. Jordan's lecture note [[Here](#)]

- In the Bayesian framework, marginal likelihoods have a natural built-in penalty for more complex models.

*Example: Remarks

$$f(x | H_i) = \int f(x | \theta) g_i(\theta) d\theta$$

- ▶ Why the marginal likelihood begins to decrease?
- ▶ The intuition is that as the complexity of the model increases, the prior $g_i(\theta)$ will be spread out across too many models and will place little prior mass on the good models.
- ▶ As a result, it causes the marginal likelihood to decrease.

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Bayesian cross-validation

Motivation

- ▶ Suppose we have different models for the same problem/data at hand.
- ▶ In this case, comparing different models can be done by a hypothesis test as discussed.
- ▶ There are different methods to select a good model. An alternative is to measure a model's predictive accuracy.
- ▶ How to measure a model's predictive accuracy in a Bayesian perspective?

Bayesian cross-validation: idea

- ▶ Let y and $\tilde{y}$ be the observed and future data, respectively.
- ▶ An ideal measure of a model's fit would be the **out-of-sample log predictive density (lpd)**:

$$\text{lpd} = \log p(\tilde{y} | y) = \log \left\{ \int p(\tilde{y} | \theta) \pi(\theta | y) d\theta \right\}.$$

- ▶ But the future data $\tilde{y}$ is unknown in general.

Bayesian cross-validation: idea

- ▶ So we may take the expected value for the lpd, which is the **expected out-of-sample log predictive density (elpd)**:

$$\text{elpd} = \mathbb{E}_0 \log p(\tilde{y} | y) = \int \log p(\tilde{y} | y) p_0(\tilde{y}) d\tilde{y},$$

where p_0 is true density/model.

- ▶ If we have samples of future data, say $\{\tilde{y}^1, \dots, \tilde{y}^M\}$, we can use the Monte Carlo approximation:

$$\int \log p(\tilde{y} | y) p_0(\tilde{y}) d\tilde{y} \approx \frac{1}{M} \sum_{\nu=1}^M \log p(\tilde{y}^\nu | y),$$

which corresponds to the **Bayesian cross-validation** (CV) (Gelman et al., 2014).

Bayesian cross-validation

- ▶ In the Bayesian CV, at first, we repeatedly divide the data (n_{cv} times) into a training set y_{train} and a test set y_{test} .
- ▶ Let $I_1(\nu)$ and $I_2(\nu)$ be indices for the ν th training set and test set, respectively, where $I_1(\nu) \cup I_2(\nu) = \{1, \dots, n\}$.
- ▶ Then, the estimated out-of-sample log predictive density is

$$\begin{aligned} \text{lpd}_{cv} &= \sum_{\nu=1}^{n_{cv}} \log p(y_{\text{test}}^{\nu} | y_{\text{train}}^{\nu}) \\ &= \sum_{\nu=1}^{n_{cv}} \log \left\{ \int p(y_{\text{test}}^{\nu} | \theta) \pi(\theta | y_{\text{train}}^{\nu}) d\theta \right\}. \end{aligned}$$

Bayesian cross-validation: remark

$$\begin{aligned}\text{lpd}_{cv} &= \sum_{\nu=1}^{n_{cv}} \log p(y_{\text{test}}^{\nu} | y_{\text{train}}^{\nu}) \\ &= \sum_{\nu=1}^{n_{cv}} \log \left\{ \int p(y_{\text{test}}^{\nu} | \theta) \pi(\theta | y_{\text{train}}^{\nu}) d\theta \right\}.\end{aligned}$$

- ▶ The above integral may not be available in a close form.
- ▶ In that case, we can again use the Monte Carlo approximation based on MCMC samples of θ .
- ▶ However, it would be computationally expensive in practice.

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Remark and next week

Remark

- ▶ In today's lecture, we explored two methods for model selection: Bayesian hypothesis testing and Bayesian cross-validation.
- ▶ Among these, Bayesian hypothesis testing can be considered as a more sophisticated criterion.
- ▶ Bayesian cross-validation can serve as a criterion for **hyperparameter selection** within a model.

Next week

Midterm exam